

Down the straight line that leads you up

X16 (2016) — English translation

An ideal polyatomic gas participates in a process, the states of which are depicted in the figure by points on the straight line $p = p_M(1 - \alpha V)$, where α is a certain constant coefficient, V is the current value of the volume. Molar heat capacity C_V does not depend on temperature.

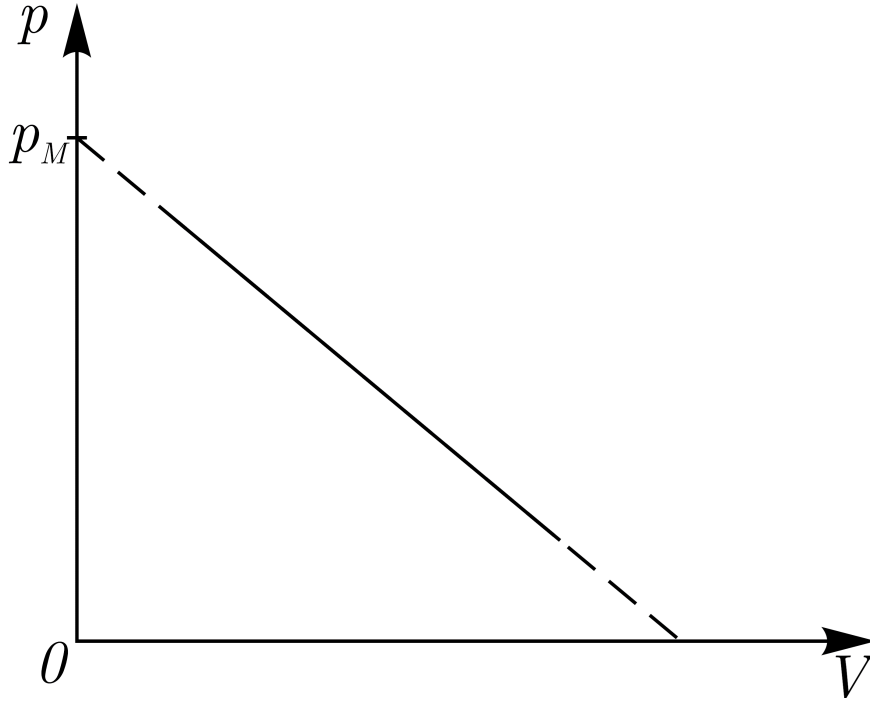


Figure 1: Figure 1.

A1	At what pressure p_0 does the heat capacity of the gas become zero?	2.00
A2	Now suppose the process involves an ideal Clausius gas, whose equation of state is $p = \frac{RT}{V - b}.$ For which gas (ideal polyatomic or ideal Clausius) is the temperature at the point of maximum heat capacity higher? Calculate the temperature difference ΔT between those points. Given: $b = 44 \text{ cm}^3/\text{mol}$, $R = 8.314 \text{ J}/(\text{mol} \cdot \text{K})$, $p_M = 1.51 \text{ MPa}$, and $b \ll V_M$, where $V_M = p_M/\alpha$ is a fixed volume.	3.00
A3	What value does the molar heat capacity C of an ideal polyatomic gas participating in the process described in paragraph A1 tend to as the volume tends to zero?	2.00

Source: <https://pho.rs/p/4003>